Predefined Time Prescribed Performance Backstepping Control for Robotic Manipulators with Input Saturation

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摘要

本文研究了具有未知动力学、有界扰动以及执行器饱和条件下的机器人机械臂轨迹跟踪控制问题，提出了一种立于任何初始状态条件的预定义时间预设性能约束的自适应反步控制方法。针对传统预设性能控制中存在的初值奇异性和误差变换非光滑问题，针对每个通道独立设计了结合多项式性能函数与误差缩放函数的预定义时间误差变换结构，确保系统状态在预定义时间内严格且平滑地满足全局预设性能约束。为处理反步控制及执行器饱和输入引起的计算复杂度爆炸问题，设计了一种预定义时间饱和补偿器，以同时消除命令滤波与输入饱和的不利影响。此外，引入一阶滑模扰动观测器实时估计并补偿系统中未知的有界扰动，利用径向基函数神经网络对系统未建模动态进行辨识和补偿，从而提高了控制系统的鲁棒性和跟踪精度。同时，结合预定义时间理论和自适应动态障碍李雅普诺夫函数，严格证明了闭环系统在预定义时间内的全局稳定性和误差收敛性。数值和实验结果结果验证了所提出方法的有效性，与现有方法相比，本文方法在响应速度、稳态精度及鲁棒性能方面表现出明显优势。

Keywords: Prescribed performance control, predefined-time stability, adaptive dynamic barrier Lyapunov function, backstepping control, input saturation, robotic manipulators' trajectory tracking.

1 Global prescribed–performance function

2 Global prescribed–performance tunnel with asymmetric boundaries (Scheme A)

Let $T_p > 0$ be the predefined convergence time and $0 < p \le 1$. Define the normalized time

$$b(t) = \left(\frac{t}{T_p}\right)^p \in (0, 1], \qquad s(b) = 1 - 3b^2 + 2b^3, \quad s(0) = 1,\ s(1) = 0,\ s'(1) = 0, \tag{1}$$

and the kernel $\phi(b) = -\ln b$ so that $\phi(b) \to +\infty$ as $b \to 0^+$ and $\phi(1) = 0$.

For each channel $i = 1, \ldots, n$, define the *symmetric base performance radius*

$$\rho_i(t) = \begin{cases} a + \sigma_i(t)\, \phi\big(b(t)\big)\, s\big(b(t)\big), & 0 < t < T_p, \\ a\Big[1 + g(t)\, \Sigma_i(t)\Big], & t \ge T_p, \end{cases} \tag{2}$$

where $a > 0$ is the steady-state accuracy, and

$$\sigma_i(t) =_{[\sigma_{\min}, \sigma_{\max}]} \Big(\sigma_0 + k_u r_{u,i} + k_d r_{d,i} + k_e r_{e,i}\Big), \tag{3}$$

$$\tau_u \dot r_{u,i} = -r_{u,i} + |\Delta u_i|, \quad \tau_d \dot r_{d,i} = -r_{d,i} + |d_i|, \quad \tau_e \dot r_{e,i} = -r_{e,i} + |e_i|, \tag{4}$$

$$g(t) = 1 - \exp\!\big(-\kappa (t - T_p)^2\big), \qquad \kappa > 0,\ g(T_p) = g'(T_p) = 0, \tag{5}$$

$$\Sigma_i(t) =_{[0,\,\Sigma_{\max}]} \Big(k_u r_{u,i} + k_d r_{d,i} + k_e r_{e,i}\Big). \tag{6}$$

Asymmetric/tunnel envelopes

On top of the base radius $\rho_i(t)$, define the *asymmetric* upper/lower envelopes

$$\rho_i^+(t) \ = \ \rho_i(t)\Big(1 + \delta_i(t)\Big) + \beta_i(t), \qquad \rho_i^-(t) \ = \ \rho_i(t)\Big(1 - \delta_i(t)\Big) - \beta_i(t), \tag{7}$$

with $0 \le \delta_i(t) < 1$ and a signed offset $\beta_i(t)$. The tracking error constraint is enforced as

$$-\rho_i^-(t) \ < \ e_i(t) \ < \ \rho_i^+(t), \qquad \forall\, t \ge 0. \tag{8}$$

The *relative widening* $\delta_i(t)$ and the *directional bias* $\beta_i(t)$ are chosen as

$$\delta_i(t) =_{[\,0,\,\delta_{\max}\,]} \Big(\delta_0 + c_u r_{u,i} + c_d r_{d,i} + c_e r_{e,i}\Big), \tag{9}$$

$$\beta_i(t) = \beta_0\, g(t)\, \tanh\!\big(\gamma_\beta\, e_i(t)\big), \qquad \beta_0 > 0,\ \gamma_\beta > 0. \tag{10}$$

Here $\delta_i(t)$ provides tunnel opening/shaping according to saturation/disturbance/error levels; $\beta_i(t)$ yields a one-sided widening aligned with the error direction. The gate $g(t)$ guarantees a C^1 activation around $t = T_p$.

Remark (one-sided safety widening).

If the actuator saturates predominantly in one direction, replace $\beta_i(t)$ by $\beta_i(t) \cdot \frac{1+(e_i)}{2}$ (or its negative) to widen only the impacted side.

Derivatives (for BLF cancellation)

Let $\dot b = \frac{p}{T_p}(t/T_p)^{p-1}$, $\phi'(b) = -1/b$, $s'(b) = -6b + 6b^2$. For $0 < t < T_p$,

$$\dot\rho_i = \dot\sigma_i\, \phi(b)\, s(b) + \sigma_i \Big[\phi'(b)\, s(b) + \phi(b)\, s'(b)\Big]\dot b. \tag{11}$$

For $t \ge T_p$,

$$\dot\rho_i = a\Big[g'(t)\, \Sigma_i(t) + g(t)\, \dot\Sigma_i(t)\Big], \qquad \dot\Sigma_i = \Big(k_u \dot r_{u,i} + k_d \dot r_{d,i} + k_e \dot r_{e,i}\Big). \tag{12}$$

Thus,

$$\dot\rho_i^{\pm} = \dot\rho_i\big(1 \pm \delta_i\big) \ \pm \ \rho_i\, \dot\delta_i \ \pm \ \dot\beta_i, \tag{13}$$

with

$$\dot\delta_i = \big(c_u \dot r_{u,i} + c_d \dot r_{d,i} + c_e \dot r_{e,i}\big), \tag{14}$$

$$\dot\beta_i = \beta_0\Big[g'(t)\tanh(\gamma_\beta e_i) + g(t)\, \gamma_\beta\, {}^2(\gamma_\beta e_i)\, \dot e_i\Big]. \tag{15}$$

In the BLF derivative, include the terms $+\dot\rho_i^{\pm}/\rho_i^{\pm}$ to cancel boundary variation.

NMF mapping under tunnel envelopes

Define the normalized error $s_i(t) \in (0,1)$ by the affine normalization to the asymmetric interval:

$$s_i(t) = \frac{e_i(t) + \rho_i^-(t)}{\rho_i^+(t) + \rho_i^-(t)} \in (0,1), \qquad \zeta_i(t) = \ln\frac{s_i(t)}{1 - s_i(t)}. \tag{16}$$

Its derivative is

$$\dot s_i = \frac{\dot e_i + \dot\rho_i^-}{\rho_i^+ + \rho_i^-} - \frac{(e_i + \rho_i^-)\,(\dot\rho_i^+ + \dot\rho_i^-)}{(\rho_i^+ + \rho_i^-)^2}, \tag{17}$$

where $\dot\rho_i^{\pm}$ are given by (**??**). Boundedness of ζ_i implies $e_i \in (-\rho_i^-, \rho_i^+)$.

Properties

The functions $\rho_i^{\pm}(t)$ in (**??**) have the following properties.

(i) (*Feasible at* $t = 0^+$*.*) Since $\phi(b) \to +\infty$ as $b \to 0^+$, $\rho_i(0^+) = +\infty$, hence any finite $e_i(0)$ satisfies $|e_i(0)| < \rho_i(0^+)$. With δ_i, β_i bounded, $\rho_i^{\pm}(0^+) = +\infty$.

(ii) (*C^1 connection at T_p.*) Because $s(1) = 0$ and $\phi(1) = 0$, we have $\rho_i(T_p^-) = a$. Moreover, $s'(1) = 0$ and $g(T_p) = g'(T_p) = 0$ yield $\dot{\rho}_i(T_p^-) = \dot{\rho}_i(T_p^+) = 0$. Since δ_i, β_i are driven by $r_{\cdot,i}$ and $g(t)$, we also have $\dot{\rho}_i^{\pm}(T_p^-) = \dot{\rho}_i^{\pm}(T_p^+) = 0$.

(iii) (*Monotone shrinking on* $(0, T_p)$*.*) For $0 < t < T_p$, $\rho_i(t)$ is strictly decreasing and C^1 provided $\sigma_i > 0$; $\rho_i^{\pm}$ inherit this monotonicity modulo bounded δ_i, β_i.

(iv) (*Slowly varying post-T_p.*) For $t \geq T_p$, $\rho_i^{\pm}(t) = a\,[1 + g(t)\Sigma_i(t)](1 \pm \delta_i) \pm \beta_i$ remain positive by design; low-pass filters in (**??**) ensure slow variation.

Default parameters (for a visible tunnel effect)

A practical choice is $\delta_0 = 0.1$, $\delta_{\max} = 0.5$, $\beta_0 = 0.3a$, $\gamma_\beta = 3$, $\kappa = 10/T_p^2$, and $\tau_u = \tau_d = \tau_e \in [0.05, 0.1]$. Gains $k_{\cdot}, c_{\cdot}$ reflect actuator saturation severity; $\Sigma_{\max}$ and $\sigma_{\max}$ serve as safety caps.

Statements and Declarations

Funding

This work is supported by the National Key Research and Development Program of China (2023YFC3008802), the National Natural Science Foundation of China (52075465), and the Key Products of Hunan Province's Manufacturing Industry "Unveiled and Leading" Project (2023GXGG018), the Project of Scientific Research Fund of the Hunan Provincial Science and Technology Department (2023JJ50021, 2023GK2029, No.2024JC1003, No.2024JJ1008, 2024JJ2051, 2023GK2026), "Algorithmic Research on Mathematical Common Fundamentals" Program for Science and Technology Innovative Research Team in Higher Educational Institutions of Hunan Province of China, the Hunan Provincial Department of Education Excellent Youth Program (23B0162).

Conflicts of Interest

The authors declare no conflict of interest.

Author contribution

Shuli Liu: Writing-Original Draft, Methodology, Software, Visualization, Data curation. Yi Liu:Resources, Investigation, Formal analysis, Conceptualization, Writing-Review & Editing. Jingang Liu: Project administration, Validation, Funding acquisition, Writing-Review & Editing. Yin Yang: Supervision, Funding acquisition, Writing-Review & Editing. All authors have reviewed and approved the final version of the manuscript.

Data Availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.